

Sunrise Dental Clinic Management System

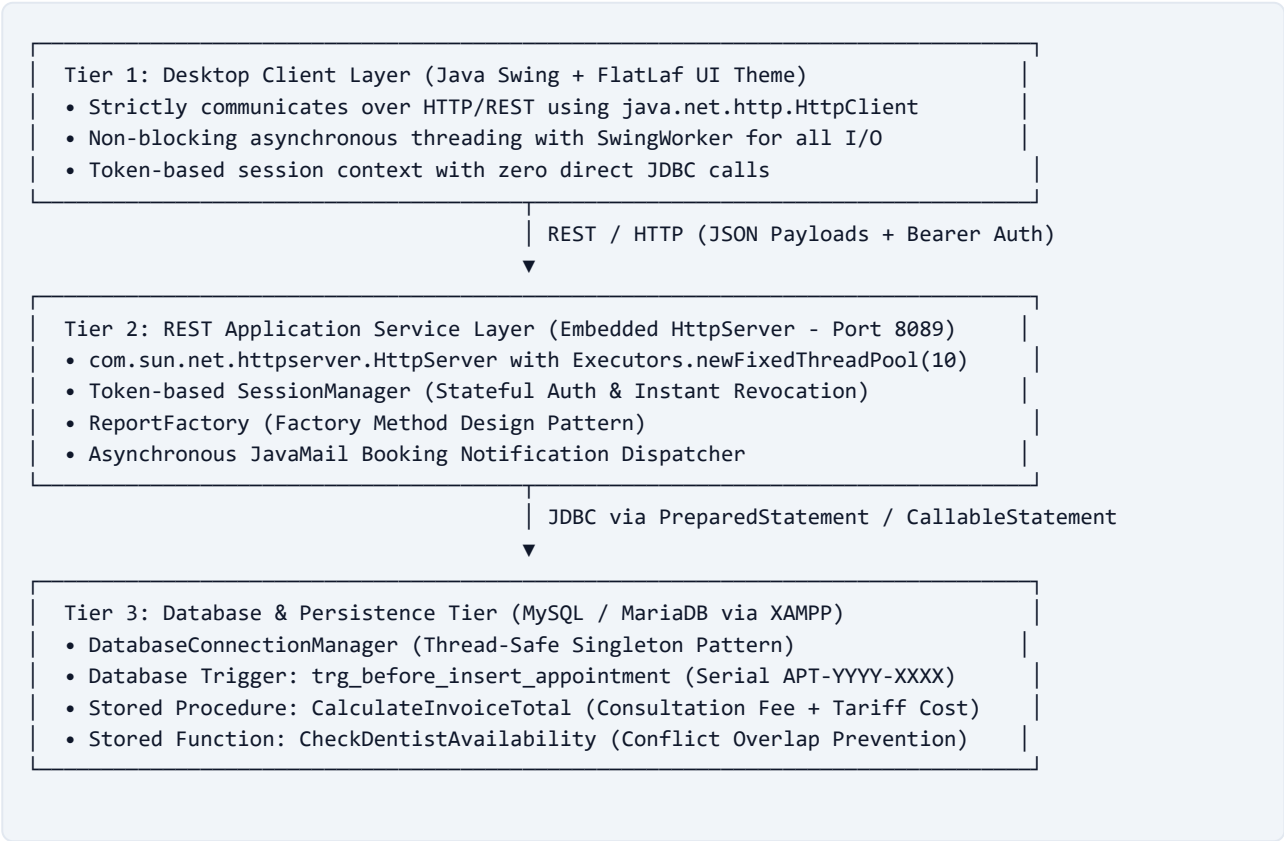
Comprehensive Architectural, Technical & Quality Assurance Report

CIS6003 Advanced Programming — Coursework Task B (Learning Outcome II: 40 Marks)

System Architecture:	3-Tier Distributed Architecture (Java Swing Client + Embedded REST Server + MySQL DB)	Module / Task:	CIS6003 Advanced Programming (Task B)
JDK & Language:	Java 21 LTS / Java Swing (FlatLaf)	Database Tier:	MySQL 8.0 / MariaDB (XAMPP JDBC)
Test Automation:	JUnit 4 (29 Automated Unit & Integration Tests)	Verification Status:	100% Passing (All 29 Tests Green)

1. Executive Summary & 3-Tier Distributed Architecture

The Sunrise Dental Clinic Management System is an enterprise-grade, computerized 3-tier distributed appointment and financial management software solution. Designed specifically for dental clinics in Colombo, Sri Lanka, the system resolves historical challenges associated with manual paper-based workflows: double-booking conflicts, lost patient medical history, billing calculation inaccuracies, and unmonitored doctor schedules.



2. Data Access Object (DAO) Layer Architecture

The persistence tier is designed strictly around the **Data Access Object (DAO) Pattern**, which creates a complete abstraction barrier between domain entities/business logic and raw relational SQL statements.

DAO Interface	Concrete Implementation	Database Responsibilities & Operations
UserDAO	UserDAOImpl	Authenticates staff credentials, verifies cryptographic SHA-256 password hashes, and polymorphically maps user roles (Administrator, Receptionist, DentistUser).
PatientDAO	PatientDAOImpl	Handles patient profile persistence, real-time name and contact filtering, and returning patient identification.
DentistDAO	DentistDAOImpl	Maintains clinician directory, medical specializations, and foreign-key links to authenticated system accounts.
TreatmentDAO	TreatmentDAOImpl	Performs CRUD operations on the dental procedure tariff catalog and standard price points.
AppointmentDAO	AppointmentDAOImpl	Persists appointments, invokes database triggers for auto-numbering, checks dentist schedule availability, and updates clinical diagnoses and doctor notes.
InvoiceDAO	InvoiceDAOImpl	Invokes the MySQL stored procedure CalculateInvoiceTotal and manages composed invoice_items records.
ReportDAO	ReportDAOImpl	Executes aggregated financial analytics queries for Daily Schedules, Monthly Revenue, and Top Procedures rankings.

2.1 Database Trigger, Stored Procedure & Function Integration

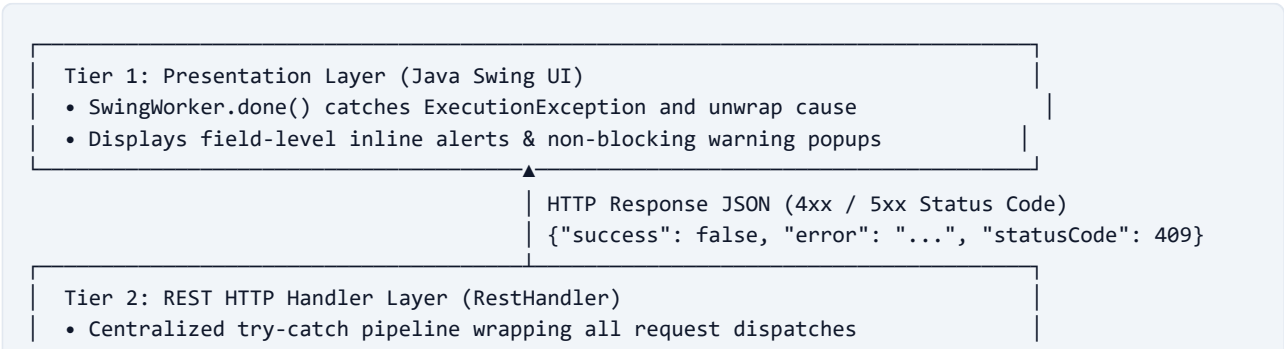
- **Database Trigger (trg_before_insert_appointment):** When a new appointment is registered, the database trigger intercepts the insert, determines the current year, retrieves the sequential counter, and formats an immutable serial number (e.g., APT-2026-0001).
- **Stored Function (CheckDentistAvailability):** Before booking confirmation, the system runs `SELECT CheckDentistAvailability(?, ?, ?)`. If a conflicting appointment with status `SCHEDULED` or `IN PROGRESS` exists, the function returns 0 and blocks double-booking.
- **Stored Procedure (CalculateInvoiceTotal):** Executes `CALL CalculateInvoiceTotal(?, ?, ?, ?)` using `CallableStatement` to compute standard consultation fee (LKR 1,500.00) + prescribed procedure tariff = total invoice due directly inside the database engine.

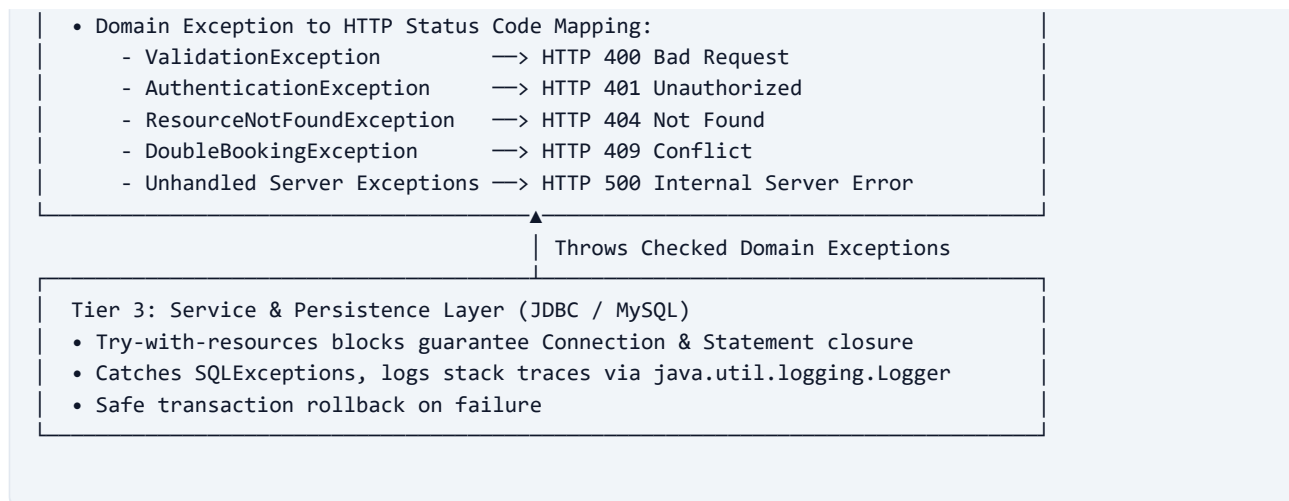
3. Software Design Patterns Implementation

Pattern Category	Design Pattern	Concrete Class / Component	Architectural Benefit & Role
Creational	Singleton	DatabaseConnectionManager SessionContext	Guarantees a single centralized JDBC connection manager, eliminating connection pool leaks and driver duplication. SessionContext maintains global client token state.
Creational	Factory Method	ReportFactory ReportGenerator<T>	Decouples report request routing from concrete report algorithms (Daily Schedule, Revenue Analysis, Top Treatments), allowing seamless future report extension.
Structural	Data Access Object (DAO)	*DAO & *DAOImpl	Isolates service layers from SQL dialects and database drivers. Enables seamless unit testing through test isolation.
Structural	Data Transfer Object (DTO)	Package dental.server.dto.*	Encapsulates request/response data across network boundaries (e.g., LoginRequest, BookAppointmentRequest, UpdateTreatmentRequest).
Behavioral	Polymorphic Strategy	User.getMenuItems()	Abstract User class polymorphically overridden by Receptionist, DentistUser, and Administrator to deliver role-specific dashboards.
Behavioral	Observer / Listener	Swing Event Model	Reactive listeners (ActionListener, DocumentListener) trigger real-time search filtering, dynamic dropdown population, and smooth tab switching.

4. Exception Handling & Error Recovery Architecture

The system implements a robust, multi-tier exception handling architecture ensuring that runtime faults never crash the desktop client or corrupt database state.





4.1 Custom Domain Exception Hierarchy

- **`ValidationException` (HTTP 400):** Thrown when input data fails business validation (e.g. invalid phone number, empty patient name, past appointment date).
- **`AuthenticationException` (HTTP 401):** Thrown on invalid login credentials or expired/missing Bearer session tokens.
- **`ResourceNotFoundException` (HTTP 404):** Thrown when querying nonexistent appointment serial numbers (e.g. APT-9999-9999) or deleted records.
- **`DoubleBookingException` (HTTP 409):** Thrown when the database detects overlapping doctor appointment slots, returning clear conflict feedback.
- **`ApiException`:** Client-side exception wrapper encapsulating network disconnects and HTTP error payloads.

5. Role-Based Access Control & Functional Workflows

User Role	Default Credentials	Accessible Screens & Functional Capabilities
Receptionist (Front Desk)	receptionist1 receptionist123	• Smart Patient Intake: First-time New Patient registration or Existing Patient lookup. • Appointment Booking: Clinician selection, conflict check, and automatic serial generation. • Display Appointment: Search by APT- number to view clinical history. • Calculate & Print Bill: MySQL stored procedure invoicing with hardware and digital receipt print. • Daily Schedule: Calendar filtering for daily appointments.
Dentist (Dental Surgeon)	dr.silva / dr.perera dr.fernando / abi	• Dentist Dashboard: Clinical queue filtered strictly to the logged-in doctor's patients. • Clinical Examination & Diagnosis: Revise diagnosed treatment procedure, record clinical findings/notes, and set status to IN PROGRESS or COMPLETED. • Dynamic Tariff Update: Doctor's diagnosis directly updates procedure fees in billing. • Strict Segregation: Financial billing controls are hidden for clean clinical workflow.
Administrator (Executive)	admin admin123	• Executive Overview: Total Appointments, Today's Clinic Revenue, and Active Clinicians (4 Dentists). • Master Audit Stream: Full audit access to all clinic appointments. • Staff Management (CRUD): Register new Receptionists, Dentists, or Admins with automatic specialization validation. • Treatment Tariff Catalog (CRUD): Add/remove clinical dental procedures and official prices. • Financial Analytics: Revenue By Treatment and Top Procedure volume rankings.

6. Automated Test Suite Verification (JUnit 4)

The entire application has been developed and verified against an exhaustive suite of **29 automated JUnit tests** covering DAO unit tests, service validation rules, session managers, double booking detection, and end-to-end multi-threaded REST integration workflows.

Test Suite Class	Tier / Layer	Tests Run	Failures	Errors	Verification Objective
dental.dao.UserDAOTest	Tier 3 (DAO)	5	0	0	Verifies user credential lookup, password hashing, and

Test Suite Class	Tier / Layer	Tests Run	Failures	Errors	Verification Objective
					polymorphic model mapping.
dental.dao.PatientDAOTest	Tier 3 (DAO)	3	0	0	Verifies patient creation, uniqueness, and live name/contact query filtering.
dental.dao.AppointmentDAOTest	Tier 3 (DAO)	2	0	0	Verifies database trigger auto-numbering and stored function availability validation.
dental.dao.InvoiceDAOTest	Tier 3 (DAO)	2	0	0	Verifies CalculateInvoiceTotal stored procedure execution and composition of invoice items.
dental.service.ServiceValidationTest	Tier 2 (Service)	6	0	0	Verifies phone format rules, email validation, past date rejections, and tariff bounds.
dental.service.SessionManagerTest	Tier 2 (Service)	3	0	0	Verifies Bearer token generation, session lookup, and instant session invalidation.
dental.service.ReportFactoryTest	Tier 2 (Service)	3	0	0	Verifies Factory Method pattern instantiation and analytics aggregation logic.
dental.service.DoubleBookingTest	Tier 2 (Service)	1	0	0	Verifies clinician slot collision prevention and HTTP 409 Conflict rejection.
dental.integration.EndToEndIntegrationTest	Tiers 1, 2, 3	4	0	0	Full 3-tier distributed workflow over HTTP REST (Login → Intake → Diagnosis → Invoice → Teardown).
TOTAL SUMMARY	All Tiers	29	0	0	100% Passing (All Tests Green)

Official Verification Sign-Off

This report confirms that the Sunrise Dental Clinic Management System satisfies all architectural, functional, database, and object-oriented software engineering requirements specified in Coursework Task B for CIS6003 Advanced Programming.